

8(a)	• force per unit length • force per unit current • length/current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
8(b)	change in potential energy = change in kinetic energy or $qV = \frac{1}{2}mv^2$	B1
	$v = \sqrt{(2qV / m)}$	A1
8(c)(i)	magnetic force = centripetal force or $Bqv = mv^2 / r$	M1
	clear substitution of expression for v and correct algebra leading to $q / m = 2V / B^2r^2$	A1
8(c)(ii)	$q / m = (2 \times 230) / [(0.38 \times 10^{-3})^2 \times 0.14^2]$	C1
	$= 1.6 \times 10^{11} \text{ C kg}^{-1}$	A1
8(c)(iii)	(for α -particle,) q / m is (much) smaller	B1
	r would be <u>much</u> larger	B1